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Lorem Ipsum University

Abstract

Equivariance to rigid-body transformations is a cornerstone of generalisation in motion planning, and learning-based planners do not inherit it. Existing work restores it in the data, in the inference operator, or in the weights. All three modify the model. None measures what the problem statement already supplies. A planning query names a start and a goal, and those two points determine a frame in closed form. Expressing the task in that frame removes five of the six degrees of freedom of $SE(3)$ exactly, at initialisation, with no constraint on the architecture. We build all three mechanisms for the sixth on a cluttered 3D benchmark, holding architecture, data and training budget fixed, and score every model on problems it never saw, beside a straight line from start to goal. The frame raises the held-out collision-free rate from $14.60 \pm 0.20\%$ to $51.10 \pm 0.25\%$, against 15.6% for the straight line, which the world-frame model does not beat. The three mechanisms for the remaining degree of freedom are worth under a point each, though the constrained architecture reaches any given level two to three times sooner. Two models that are equally equivariant differ by 28 points, so how equivariant a planner became does not predict what the symmetry bought it. We also calibrate how large the frame’s effect is on this benchmark. Giving each waypoint its distance and direction to the nearest obstacle is worth +40.0 points where the frame is worth +36.5, so the representation change studied here is not the largest effect available, and the two are not substitutes.

I. Introduction

Motion planning is a fundamental problem in robotics, requiring the generation of collision-free, kinematically feasible trajectories from a start to a goal configuration in complex environments. Equivariance to rigid-body transformations is a cornerstone of generalization in this setting. It ensures that a planner solves structurally identical tasks identically, regardless of where in the world they are posed. Sampling-based [14] and optimization-based [24, 30] algorithms are equivariant by construction, since they act on the geometry directly, and they offer completeness or convergence guarantees. Their per-query computational cost limits real-time applicability. Learning-based planners [3, 8, 12, 22] amortize that cost across queries and have emerged as a promising alternative. They do not inherit the symmetry. A network trained on world coordinates relearns the

same maneuver at every position and orientation.

Existing work restores it in one of three parts of the pipeline. In the data, through augmentation. In the inference operator, through frame averaging or canonicalization [13, 20]. Or in the weights, through architectures equivariant by construction [5, 6, 9, 26], as adopted in recent robot learning systems [23, 25, 28, 29]. All three modify the model. None measures how much of the symmetry the problem statement already supplies before any model is built.

A planning query names a start and a goal, and those two points determine a canonical frame. Place the origin at their midpoint, align the first axis with the direction between them, and express every waypoint and every obstacle in that frame. Three translational and two rotational degrees of freedom vanish exactly, at initialisation, at no computational cost and with no constraint on the architecture. The six numbers that conditioned the network on the query collapse to one invariant scalar. The frame needs a second axis to be determined, and ours takes it from a fixed world axis, so rotating the whole problem rolls the frame rather than leaving it unchanged. What survives is exactly that rotation: $SE(3)$ collapses to $SO(2)$. No continuous rule can fix the remainder [17], so the sixth degree of freedom is left to one of the three mechanisms above, and we build all three. Fig. 1 takes the group apart one stage at a time.

We build all three on a cluttered 3D benchmark, hold the architecture, data and training budget fixed, and score every model on problems it never saw, alongside a straight line from start to goal and three classical planners. The frame is worth +36.5 points of held-out collision-free rate. The three mechanisms for the sixth are worth +0.8, +0.68 and +0.15. A straight line from start to goal scores 15.6% , and the world-frame model, trained to a plateau, reaches 14.60% .

Our contributions are as follows.

- **Five of six degrees of freedom are free.** A frame derived from the query in closed form removes them exactly, raising the held-out collision-free rate from 14.60% to 51.10% with no change to the architecture, the data or the budget.
- **All three mechanisms for the sixth, measured against each other.** Neither of the two run to convergence recovers a point, and the third was already not significant at a fixed budget. The constrained architecture reaches any given level two to three times sooner, so what the symmetry buys is optimisation rather than accuracy. We explain the null re-

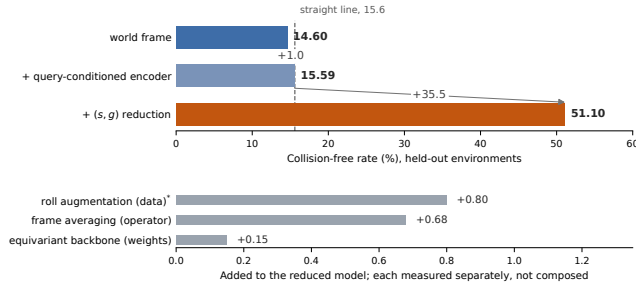


Figure 1: **Sixty environments, trained to convergence.** *Top:* a frame read off the query in closed form is worth +35.5 points over the bar immediately above it, and +36.5 over the plain world frame. Showing the obstacle encoder the same query without changing frame is worth +1.0. The dashed line is a straight segment from start to goal on the identical problems, which the world-frame model does not beat. *Bottom:* the three mechanisms for the one degree of freedom the frame cannot remove. Each is measured against its own baseline and they are alternatives rather than a decomposition, so they are not composed here and were never composed in the experiments. Two of the three are converged measurements. Roll augmentation, starred, is the 20-epoch grid, where it was already not significant across three seeds and was not run further.

sult rather than inferring it from flat curves, by measuring the non-equivariance residual of the learned field and proving it bounds what any post-hoc symmetrisation can remove. We then show the residual does not predict what a mechanism is worth: two models that are equally equivariant differ by 28 points.

- **Calibration in both directions, including against ourselves.** Supplying each waypoint with local geometry is worth +40.0 points where the frame is worth +36.5, so the representation change studied here is not the largest effect available on this benchmark. It is not a substitute for that information either: with local geometry present, the frame still adds +24.8.

II. Related Work

Classical planners. Sampling-based [14] and optimization-based [24, 30] planners act directly on geometry, so a rigid transformation of the problem transforms the solution. They offer probabilistic completeness or local convergence, at a per-query cost that follows from searching or optimizing at inference time. We use RRT-Connect and CHOMP to generate expert trajectories, and all three to bound what is achievable on our benchmark.

Learning-based planners. MPNet [22] and Motion Policy Networks [8] amortize planning into a network evaluated in milliseconds. Diffuser [12], Motion Planning Diffusion [3] and Diffusion Policy [4] generate whole trajectories with a denoising model, and flow matching [1, 15, 16] replaces the denoising chain with a learned velocity field, which is the objective we use. None of them inherits the symmetry that the classical planners get from the geometry for free.

Restoring the symmetry. The general framing is geometric [2], and equivariant models are known to carry a strict generalisation benefit under the right conditions [7]. Group-equivariant convolution [5], tensor field networks [26], SE(3)-Transformers [9] and vector neurons [6] build the constraint into the weights, packaged for practitioners by [10], and robot learning has adopted them widely [23, 25, 27, 28, 29]. Frame averaging [20] and learned canonicalization [13] act at inference instead, and the projection property our bound rests on is theirs. The Lie derivative [11] measures how equivariant a trained model became. That is the quantity we bound, and it is not the quantity, we find, that predicts anything. Frames read off the start and the goal are common practice in manipulation policy learning, where they are used and not measured. The construction is not what we claim. What it is worth, against a floor and against the alternatives, is.

III. Method

In this section, we begin by formally defining the generative motion planning objective and the symmetry it carries (Sec. III-A). We then decompose SE(3) into the part a planning query determines and the part it does not, and state precisely what survives that reduction (Sec. III-B). To establish that the remainder cannot be removed by the same route, we give two obstructions, one topological and one that applies to scenes carrying a symmetry of their own (Sec. III-C). Finally, we enumerate the three places a symmetry can be imposed, in the training data, in the inference operator, and in the weights (Sec. III-D), and define the non-equivariance residual that bounds what the second of them can remove.

III-A. Problem formulation

We formulate motion planning in the workspace $\mathbb{R}^3$. A task is the tuple $\mathcal{P} = (\mathbf{s}, \mathbf{g}, \mathcal{O})$, where $\mathbf{s}, \mathbf{g} \in \mathcal{C}$ are the start and goal configurations, $\mathcal{C} = \mathbb{R}^3$ for the point mass studied throughout and $\mathcal{C} = \text{SE}(3)$ for the rigid-body domain of Sec. V-D, and $\mathcal{O}$ is a set of convex obstacles inducing a signed distance field $d_{\mathcal{O}} : \mathbb{R}^3 \rightarrow \mathbb{R}$. For $\mathcal{C} = \text{SE}(3)$ we overload $d_{\mathcal{O}}(p)$ to mean the minimum, over the spheres composing the body at pose p , of the field at each centre less that sphere’s radius, so ρ is absorbed into the body and the constraint below reads unchanged. A

trajectory is a waypoint sequence $x = [p_1, \dots, p_N] \in \mathcal{C}^N$ and $\gamma(x)$ the path it traces, piecewise linear in position and geodesic in orientation. The feasible set of $\mathcal{P}$ is

$$\mathcal{S}(\mathcal{P}) = \{x : p_1 = \mathbf{s}, p_N = \mathbf{g}, d_{\mathcal{O}}(p) \geq \rho \ \forall p \in \gamma(x)\}, \quad (1)$$

with ρ the robot radius. The objective is to learn a generative model $p_{\theta}(x | \mathcal{P})$ supported on $\mathcal{S}(\mathcal{P})$.

A rigid transformation $T \in \text{SE}(3)$ acts on a task by $T\mathcal{P} = (T\mathbf{s}, T\mathbf{g}, T\mathcal{O})$ and on a trajectory waypointwise, $(Tx)_i = Tp_i$. The task is then exactly $\text{SE}(3)$ -equivariant:

$$\mathcal{S}(T\mathcal{P}) = T\mathcal{S}(\mathcal{P}), \quad \forall T \in \text{SE}(3), \quad (2)$$

and a generative planner should inherit this,

$$p_{\theta}(Tx | T\mathcal{P}) = p_{\theta}(x | \mathcal{P}), \quad \forall T \in \text{SE}(3). \quad (3)$$

We use conditional flow matching [15], which learns a time-dependent velocity field $v_{\theta}(x_t, t, \mathcal{P})$ whose integral transports a simple prior to $p_{\theta}(x | \mathcal{P})$. Velocities are free vectors, so the corresponding condition on the field is

$$v_{\theta}(Tx_t, t, T\mathcal{P}) = Rv_{\theta}(x_t, t, \mathcal{P}), \quad T = (R, \tau) \in \text{SE}(3), \quad (4)$$

which yields Eq. (3) whenever the prior transported by the flow is itself invariant under the action. Nothing in the flow matching objective enforces either one, and Sec. V-E measures how far a trained field falls from Eq. (4).

III-B. Decomposing the group

The motion planning task gives us a canonical frame at initialisation, and at no cost. Unfortunately the canonicalization is not entirely complete, since fixing the rotation about the first axis needs a second reference axis, and the query does not give us one. We decompose as follows.

We declare $m = \frac{1}{2}(\mathbf{s} + \mathbf{g})$, $\ell = \|\mathbf{g} - \mathbf{s}\| > 0$ and $\hat{u} = (\mathbf{g} - \mathbf{s})/\ell$. The second axis is not determined by the query, and ours is read off the world basis vector e_k that $\hat{u}$ is least aligned with,

$$k = \arg \min_i |\hat{u}_i|, \quad \hat{n} = \frac{\hat{u} \times e_k}{\|\hat{u} \times e_k\|}. \quad (5)$$

The denominator never vanishes: k forces $\hat{u}_k^2 \leq \frac{1}{3}$, so $\|\hat{u} \times e_k\|^2 = 1 - \hat{u}_k^2 \geq \frac{2}{3}$. Let $R_{\mathcal{P}} \in \text{SO}(3)$ be the rotation with $R_{\mathcal{P}}^{\top} = [\hat{u} \ \hat{n} \ \hat{u} \times \hat{n}]$. The reduction is the rigid map

$$\Phi_{\mathcal{P}}(p) = R_{\mathcal{P}}(p - m), \quad (6)$$

under which the endpoints stop being inputs and become constants,

$$\Phi_{\mathcal{P}}(\mathbf{s}) = (-\frac{\ell}{2}, 0, 0), \quad \Phi_{\mathcal{P}}(\mathbf{g}) = (+\frac{\ell}{2}, 0, 0). \quad (7)$$

Hence the network never sees a world coordinate. The task is carried into the frame, $\hat{x}_t = \Phi_{\mathcal{P}}(x_t)$ and $\hat{\mathcal{O}} =$

$\Phi_{\mathcal{P}}\mathcal{O}$, obstacle centres transforming as points, box edges as free vectors under $R_{\mathcal{P}}$ alone, and radii not at all. The field is evaluated entirely inside the frame, $\hat{v} = v_{\theta}(\hat{x}_t, t | \ell, \hat{\mathcal{O}})$, where by Eq. (7) the query enters as the single scalar ℓ in place of the six numbers of $(\mathbf{s}, \mathbf{g})$. The output is returned to the world by the rotation alone, $v = R_{\mathcal{P}}^{\top} \hat{v}$, velocities being free vectors.

Eq. (6) costs one cross product per query, constrains nothing about v_{θ} , and holds at initialisation. It fixes five of the six degrees of freedom of $\text{SE}(3)$, three of translation through m and two of rotation through $\hat{u}$. It does not fix the sixth, and Eq. (5) reads the world basis, so it turns with the world.

Proposition 1 (The residual is $\text{SO}(2)$). *Let $R_1(\theta)$ denote the rotation by θ about the first coordinate axis. For every task $\mathcal{P}$ with $\ell > 0$ and every $T \in \text{SE}(3)$ there is a $\theta = \theta(T, \mathcal{P})$ with*

$$\Phi_{T\mathcal{P}} \circ T = R_1(\theta) \circ \Phi_{\mathcal{P}}. \quad (8)$$

Proof. $\|T\mathbf{g} - T\mathbf{s}\| = \ell$, so by Eq. (7) both $\Phi_{T\mathcal{P}} \circ T$ and $\Phi_{\mathcal{P}}$ carry $\mathbf{s}$ to $(-\frac{\ell}{2}, 0, 0)$ and $\mathbf{g}$ to $(+\frac{\ell}{2}, 0, 0)$. Their composite $Q = \Phi_{T\mathcal{P}} \circ T \circ \Phi_{\mathcal{P}}^{-1}$ is therefore an orientation-preserving isometry fixing two distinct points, hence a rotation about the line joining them, which is the first coordinate axis. $\square$

Nothing in the proof refers to Eq. (5). The residual is $\text{SO}(2)$ for every rule selecting the second axis, and the rule decides only which θ . What Eq. (3) still asks of a planner in this frame is equivariance to $\text{SO}(2)$, then, and not to nothing. One rule would escape it, a rule continuous in $\mathcal{P}$ and returning the same $\hat{n}$ for $\mathcal{P}$ and $T\mathcal{P}$. Ours is not continuous: it jumps where the minimising k in Eq. (5) ties. Sec. III-C is why no rule does better.

III-C. Why no continuous rule removes the remainder

Sec. III-B removed five degrees of freedom by reading a frame off the query. Whether the sixth yields to the same treatment, under some better choice of second axis, is not a question about our construction. Two obstructions answer it, one topological and one that applies to an individual scene. Neither is original to us. We state them because the rest of the paper is an empirical answer to what they leave open.

Proposition 2 (Topological obstruction). *There is no continuous $\hat{n} : S^2 \rightarrow S^2$ with $\hat{n}(\hat{u}) \perp \hat{u}$ for every $\hat{u} \in S^2$. Every rule selecting the second axis, Eq. (5) among them, is therefore discontinuous somewhere on the sphere of query directions.*

Proof. Such an $\hat{n}$ is a continuous nowhere-vanishing tangent vector field on S^2 , which the hairy ball theorem forbids [17]. $\square$

Proposition 3 (Scenes with their own symmetry). *Let $\hat{\mathcal{O}}$ be a reduced scene with $R_1(\pi)\hat{\mathcal{O}} = \hat{\mathcal{O}}$. Then no rule reading only the reduced task can distinguish the frame $R_{\mathcal{P}}$ from $R_1(\pi)R_{\mathcal{P}}$.*

Proof. $R_1(\pi)$ fixes both endpoints of Eq. (7) and fixes $\hat{\mathcal{O}}$ by hypothesis, so the two frames present the rule with identical arguments. $\square$

Neither proposition rules out a discontinuous rule that reads the scene [13], and outside the measure-zero set of Prop. 3 such a rule can fix the roll. What we show is that the remainder cannot be removed the way the other five degrees of freedom were, in closed form from the query and continuously. Sec. V measures the three mechanisms that remain, and none of them recovers a point.

III-D. Three ways to impose the symmetry

A symmetry that the problem has and the model lacks can be imposed in three places. In the data, by training on randomly transformed copies of each problem, which teaches the symmetry without guaranteeing it. In the inference operator, by averaging the learned field over the group, which guarantees it for the averaged field without changing what was learned. In the weights, by restricting the hypothesis class to fields equivariant by construction, which guarantees it at initialisation and constrains optimisation throughout.

Averaging a field f over the K rolls $\theta_k = 2\pi k/K$ gives

$$(\mathcal{A}f)(\hat{x}) = \frac{1}{K} \sum_{k=0}^{K-1} R_1(\theta_k)^\top f(R_1(\theta_k)\hat{x}), \quad (9)$$

the orthogonal projection of f onto the fields equivariant to the cyclic subgroup $C_K \subset \text{SO}(2)$ [20].

The first two mechanisms act on a field that has already been trained, and how far such a field sits from equivariant is measurable rather than assumed. We write

$$r = \frac{\|f - \mathcal{A}f\|}{\|\mathcal{A}f\|} \quad (10)$$

for the non-equivariance residual of f , which costs one forward pass per rotation and no retraining. It bounds what the second mechanism can do.

Proposition 4 (Symmetrisation budget). *Let f^* be equivariant and $\mathcal{A}$ the projection of Eq. (9). Then*

$$\|f - f^*\|^2 = \|\mathcal{A}f - f^*\|^2 + r^2 \|\mathcal{A}f\|^2. \quad (11)$$

The squared error that any equivariantisation of f can remove is at most $r^2 \|\mathcal{A}f\|^2$, and $\mathcal{A}$ attains it. The remainder $\|\mathcal{A}f - f^\|$ is equivariant error, which no symmetrisation of f reduces. Every map carrying f to an equivariant field moves it by at least $r\|f\|/\sqrt{1+r^2}$, with equality only for $\mathcal{A}$.*

Proof. $\mathcal{A}$ is an orthogonal projection, so $f - \mathcal{A}f$ is orthogonal to the equivariant subspace, which contains $\mathcal{A}f - f^*$. $\square$

IV. Setup and Calibration

IV-A. Benchmark and protocol

PointMass3D places forty obstacles, twenty spheres and twenty oriented boxes, in $[-1, 1]^3$. The robot is a sphere of radius $\rho = 0.03$. Collision checking uses an analytic signed distance field, and $\gamma(x)$ is resampled at a spacing of 0.01 before it is tested rather than evaluated at the 64 waypoints alone. Since $d_{\mathcal{O}}$ is 1-Lipschitz, every point of $\gamma(x)$ lies within 0.005 of a tested sample, so the reported rates differ from the continuous ones by at most a 0.005 perturbation of ρ , a sixth of the robot radius. A trajectory is $N = 64$ waypoints in $\mathbb{R}^3$.

Expert paths come from RRT-Connect, then random shortcutting, uniform resampling, and CHOMP refinement. The dataset is 300 environments, each with 600 start-goal pairs and 30 paths per pair including time reversals. Environments 250 to 299 are held out and never trained on. Unless stated otherwise, models train on the first 60 of the remaining 250; the scaling study in Sec. V-A varies that count.

The planner is a conditional flow-matching model with 2.16M parameters: a dilated temporal convolutional trunk [18] with FiLM conditioning [19] and a PointNet-style obstacle encoder [21]. The encoder max-pools all forty obstacles into a single 128-dimensional vector, and FiLM applies that vector identically to every one of the 64 waypoints.

Every model is evaluated on the same 500 problems: 50 unseen environments, 10 distinct start-goal pairs each, 20 samples per pair, eight Euler steps. We report two metrics and never conflate them. A sample is *free* if the whole path is collision-free. *Best-of-20* asks whether any of the twenty samples for a query is free. They differ by about thirty points. Uncertainty is quoted across seeds for claims about a method. For comparisons against a fixed baseline measured on the same problems we use a paired bootstrap over environments.

IV-B. The floor, and a paired test against it

A learned planner is worth what it adds to the cheapest thing that solves the same problems. Joining start to goal with a straight segment is collision-free on 15.6% of the 500 held-out problems, so the clutter is dense and not so dense that the direct path is usually blocked. The world-frame model converges to $14.60 \pm 0.20\%$.

Quoting the across-seed spread here would answer a different question. That spread describes variation between training runs, not the uncertainty of a comparison

Planner	Free (%)	Clearance	s/query
straight line	15.6	0.055	<0.001
TrajOpt	74.0	0.026	0.352
CHOMP	88.8	0.019	0.394
RRT-Connect	99.6	0.017	0.284
expert (RRT+CHOMP)	100.0	0.037	0.342
flow, world frame	14.6	-0.071	0.053
flow, (s, g) reduction	51.1	-0.002	0.063

Table 1: The same 500 unseen problems solved five other ways. The learned rows are converged, three-seed means at 60 environments. Classical timings are one CPU core and one query; the learned rows are one GPU and twenty samples, so the time column is not like-for-like. Clearance is $d_{\mathcal{O}} - \rho$, the margin by which the robot misses the nearest obstacle, minimised along the densely resampled path; it is negative exactly when the path collides. Classical rows average it over solved problems only, learned rows over all samples, which is why the straight line reads +0.055 while failing 84.4% of them. The expert row is the filtered training distribution rather than a planner that never fails.

against a fixed baseline measured on the identical problems. Both quantities track per-environment clutter, so we resample environments and recompute the difference within each replicate. The paired bootstrap gives -1.02 points, a 95% interval of $[-3.06, +0.90]$, and a standard error of 1.01, less than half the 2.2 obtained by resampling either rate alone, and still spanning zero. What we defend is therefore that training on world coordinates does not beat joining the endpoints with a line, and not that it loses to one. At 20 epochs even that was unavailable (15.37 ± 0.36). Optimisation is what makes the comparison resolvable, not what creates the deficit.

None of this is what the planner is worth to a practitioner. Behind a collision check, best-of-20, the reduced model solves 87.2% of held-out queries at 60 environments against 15.8% for the best untrained prior. That is the number to read if the question is whether the sampler is useful rather than what the representation buys, subject to Table 1.

V. Results

V-A. Scaling

Across a $12.5\times$ increase in training environments the world-frame model goes from 12.9 to 15.4. The reduction goes from 19.2 to 45.5, and has not plateaued. Seed-matched, the difference grows monotonically at every size: $+6.37 \pm 0.19$, $+20.87 \pm 1.56$, $+25.85 \pm 0.37$ and $+30.08 \pm 0.38$.

The world-frame model is not short of data. Training it on random SE(3) motions of the whole problem, which is the data mechanism of Sec. III-D, moves it by at most 2.1 points and leaves it at 17.5. These cells hold the budget at 20 epochs so that representation is the only

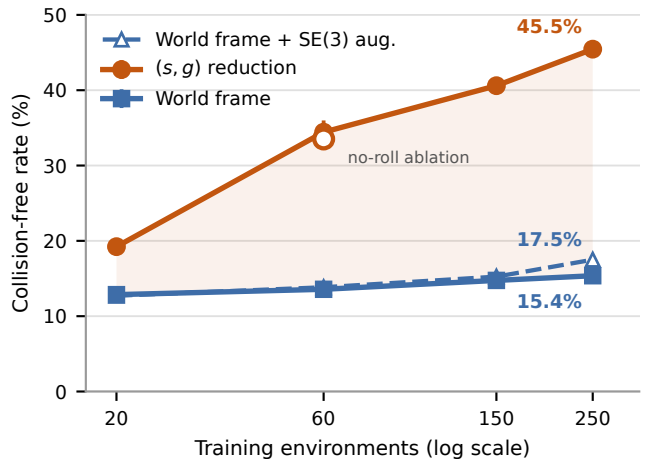


Figure 2: **Held-out collision-free rate against training-set size, at a fixed budget of 20 epochs.** The world-frame model moves 2.5 points across a $12.5\times$ increase in environments, from 12.9 to 15.4, while the reduction runs from 19.2 to 45.5 without plateauing. Holding the budget fixed is what makes representation the only variable, and it is also why these are not convergence levels: trained out, the 250-environment reduced model reaches 55.8, above the 51.10 quoted at 60 environments elsewhere. Every world-frame and reduction cell is a mean over three seeds.

thing varying, which is why 45.5 at 250 environments sits below the 51.10 that Sec. V-C reports at 60. Trained out, the 250-environment model reaches 55.8.

V-B. Control: a query-conditioned encoder

The gap admits a deflationary reading. In the world frame the obstacle encoder sees obstacles and nothing else, so its output is constant across every query in an environment, at a measured within-environment standard deviation of 0.000000. That is a property of the architecture rather than of training. Under the reduction the same encoder sees those obstacles expressed in the query’s own frame, so its code varies with the query. On this reading the +36.5 measures letting the encoder see the query, and the change of frame is incidental.

We built the model that separates the two. It stays in the world frame throughout, and the raw query is concatenated to every obstacle before the pool, so the scene code depends on the query with no change of frame anywhere. It carries 0.07% more parameters.

It converges to $15.59 \pm 0.38\%$ against the world frame’s 14.60 ± 0.20 , seed-matched +1.01 and positive on all three seeds. Query dependence is worth about one point of the 36.5. The level it reaches is the more useful number. At 15.59 it lands on the straight line’s 15.6, so showing the encoder the query is enough to reach the floor and not enough to pass it.

Mechanism	20 ep.	conv.	
(s, g) reduction	+30.1	+36.5	$n=3$
3 translations	+12.3	—	recentring
2 rotations	+18.0	—	axis
			alignment
SE(3) augmentation	+2.1	—	7% of the above
roll augmentation	+0.8	—	n.s., $n=3$
frame averaging ($K:1 \rightarrow 3$)	+0.1	+0.68	± 0.13 , $n=3$
equivariant backbone	—	+0.15	$n=3$

Table 2: Points on the held-out collision-free rate, each measured against the model without that mechanism. The rows are not additive and were never composed. The two degree-of-freedom rows were measured only on the fixed-budget grid, which is why that column exists.

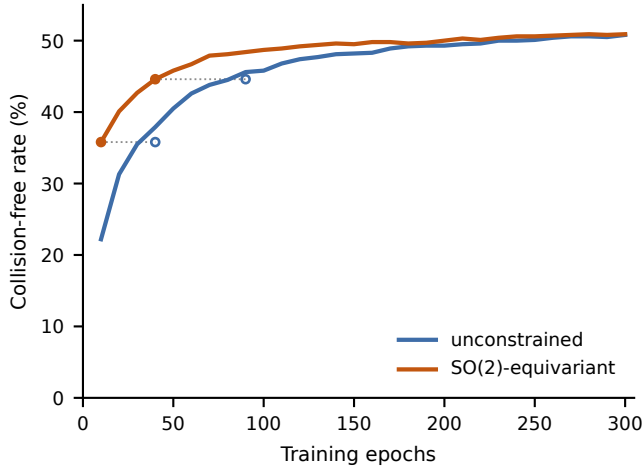


Figure 3: **Held-out collision-free rate against training epochs, 60 environments.** The constrained backbone reaches any given level two to three times sooner than the unconstrained model in the same frame, and the two converge to within 0.15 points. Seed-matched over three seeds.

V-C. Control: local geometry, and how it scales

One scene code applied identically to all 64 waypoints leaves a waypoint no channel through which to ask what is near it, and collision avoidance is that question. Sec. V-B shows the reduction addresses this indirectly. It can also be addressed directly. We append to each waypoint the signed distance to the scene and its gradient there, which for the constrained backbone splits by irreducible representation: $d_{\mathcal{O}}$ and the axial component of $\nabla d_{\mathcal{O}}$ are invariant, the normal component rotates. This is a featurisation and not an oracle, since the signed distance is a deterministic function of primitives the network already receives. What it assumes is that those primitives are available at inference, which holds on this benchmark and fails for a planner operating on perception.

Table 3 crosses the two. Local geometry is worth

60 envs, converged	global only	+ local	geometry adds
world frame	14.60	54.54	+40.0
(s, g) reduction	51.10	79.37	+28.3
+ equivariant backbone	51.27	82.13	+30.9
<i>the frame contributes</i>	+36.5	+24.8	

Table 3: Local geometry against the query frame. Every cell is a mean over two or three seeds. The two local-geometry columns are the least seed-sensitive measurements in this work, with across-seed standard errors of 0.01 and 0.10 against 0.20 and 0.25 without. The equivariant row is a seed-matched pair, 51.27 against 51.12 on the same three seeds, so the +0.15 of Table 2 is that difference and not 51.27 minus the three-seed mean 51.10 above it.

+40.0 points in the world frame, more than the frame itself is worth, so the representation change studied here is not the largest effect available on this benchmark. The two are not substitutes either: with local geometry supplied, the frame still adds +24.8. They are sub-additive. Treating them as independent predicts $14.60 + 36.5 + 40.0 = 91.1$ and the measured value is 79.37, because both address the same difficulty by different routes.

What the frame is worth beside local geometry decays with data. Differencing within seed at a matched 80-epoch budget, it is $+22.37 \pm 0.46$ points at 60 environments and $+17.24 \pm 0.17$ at 250, two seeds at each scale. The 5.1-point decay is an order of magnitude larger than either standard error. Local geometry is what scales over that range, gaining 11 points while the frame beside it loses 5.

V-D. Replications

Three things vary, one at a time. A second state space: an L-shaped rigid body in the same clutter, whose state is a full pose rather than a point, so the points-versus-vectors distinction moves inside the trajectory. Its trivial baseline is geodesic interpolation, straight in position and slerp in rotation. The world-frame model never clears that floor at either budget; the reduction clears it and is still climbing. A second generative objective: replacing flow matching with DDPM at matched network evaluations leaves the gap at $+37.01$ against $+36.50$, so the effect does not belong to the sampler. A range of clutter: on freshly generated layouts at densities neither model was trained on, the gap runs from $+29.4$ at eight obstacles to $+41.2$ at twenty-four.

The rigid-body cells are the weakest measurements in this paper. One seed, and both models below their own floor at 20 epochs. What replicates there is the ratio and not the level: at matched budgets the reduced model is $2.9\times$ the world-frame model in both domains.

Replication	Condition	World	Reduced	Gap
Second state space. SE(3) rigid body, floor 4.8				
	20 ep.	2.8	6.2	+3.4
	155/159 ep.	3.1	10.0	+6.9
Other objective. DDPM, matched network evaluations				
	20 ep.	12.27	28.97	+16.71
	300 ep.	13.51	50.52	+ 37.01
Other clutter. Obstacles per environment				
	8	61.4	90.8	+29.4
	24	24.8	66.0	+ 41.2
	40 (trained)	13.5	46.4	+32.9

Table 4: The effect survives a change of state space, a change of generative objective, and a $5\times$ range of clutter. The rigid body is an L-shaped union of five spheres whose state is a full pose, chosen to have trivial symmetry so the domain is not secretly easier; both models are below their own floor at 20 epochs and only the reduced one clears it. The second column is each model’s own best-validation checkpoint, epoch 155 for the world frame and 159 for the reduction, so neither had stopped improving when the runs ended. The density rows are freshly generated layouts at densities neither model was trained on. Single seed except the DDPM 20-epoch row.

V-E. The residual, and what it bounds

Proposition 4 bounds what symmetrising a trained field can remove. For the reduced model $r = 0.017$, so that bound is under 3×10^{-4} of the field’s squared magnitude, while the model is failing on 48.9% of its samples. The two quantities are not the same order, and no rearrangement of the symmetry closes the gap. The measurement costs one forward pass and no retraining.

What r does not do is forecast what a mechanism is worth. Measured under the full SE(3) action, which makes it defined for the world-frame model as well, augmentation drives r from 0.0881 to 0.0181 and buys 2.1 points. The reduction, at 0.0174, buys 30.1. Both residuals are single measurements on one checkpoint each, since r is a property of a particular trained field, and they differ by less than the gap between the $K=9$ and $K=32$ estimates of either.

VI. Limitations

- **The group must act on the state space, not just the workspace.** This holds for a point mass and for the rigid body of Table 4. It fails for a manipulator planning in joint space: the query still defines a frame in the workspace, but a rigid motion of the world induces no clean action on joint angles. Mobile bases, free-flying bodies and end-effector-space planning satisfy the condition. Joint-space planning for arms does not.
- **The learned planner loses to the classical ones.** RRT-Connect solves 99.6% in under 0.3 s on one CPU core. On this benchmark one should use

RRT-Connect. What we measure is what a representation change is worth, and that is only isolable where the confounds are removable.

- **One workspace family, one clutter generator, one architecture.** The claim that the sixth degree of freedom needs no treatment relies on the training distribution supplying implicit roll diversity through many start-goal directions. A dataset with few directions per environment could reverse it.
- **Seeds are not uniform.** The 60-environment cells carry three seeds, the local-geometry cells two or three, and the 250-environment grid three. Only the converged 250-environment run and the rigid-body cells of Table 4 carry one. Every claim we lean on has at least two.
- **The frame’s value beside local geometry falls with data.** It is +22.4 points at 60 environments and +17.2 at 250 on a matched budget. We have not found where that decay stops, and a reader extrapolating to much larger datasets should expect the margin to narrow rather than hold.
- **Local geometry assumes explicit primitives at inference.** The signed distance is a deterministic function of obstacle data the network already receives, so it is a featurisation rather than an oracle, but it is unavailable to a planner working from perception.

VII. Conclusion

Five of the six degrees of freedom of SE(3) were removable exactly, at initialisation and at no cost, by a frame the planning query already supplied. That was worth +36.5 points of held-out collision-free rate, against a world-frame model that did not beat a straight line. The sixth was not removable by any continuous rule, and all three mechanisms available for it recovered under a point each. The constrained architecture was not idle: it reached any given level two to three times sooner, so the symmetry bought optimisation and not accuracy. Two models that were equally equivariant differed by 28 points, so how equivariant a model became did not predict what the mechanism was worth. Giving each waypoint its distance and direction to the nearest obstacle was worth more than the frame was, and the frame still added +24.8 points on top of it.

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The software implementation was carried out with Claude Code under the first author’s direction. The classical planner implementations, used to generate expert trajectories and to bound what is achievable, follow the published algorithms [14, 24, 30] and their public reference implementations. The problem formulation, the theoretical analysis, the experimental design and the text of this paper are the authors’ own. Every reported number is produced by the released code, whose geometric and equivariance-critical components are verified by property-based tests that run on untrained networks and include negative controls.

References

- [1] M. S. Albergo and E. Vanden-Eijnden. Building normalizing flows with stochastic interpolants. In *ICLR*, 2023.
- [2] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv:2104.13478*, 2021.
- [3] J. Carvalho, A. T. Le, M. Baierl, D. Koert, and J. Peters. Motion planning diffusion: Learning and planning of robot motions with diffusion models. In *IROS*, 2023.
- [4] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. In *RSS*, 2023.
- [5] T. S. Cohen and M. Welling. Group equivariant convolutional networks. In *ICML*, 2016.
- [6] C. Deng, O. Litany, Y. Duan, A. Poulenard, A. Tagliasacchi, and L. Guibas. Vector neurons: A general framework for $SO(3)$ -equivariant networks. In *ICCV*, 2021.
- [7] B. Elesedy and S. Zaidi. Provably strict generalisation benefit for equivariant models. In *ICML*, 2021.
- [8] A. Fishman, A. Murali, C. Eppner, B. Peele, B. Boots, and D. Fox. Motion policy networks. In *CoRL*, 2022.
- [9] F. B. Fuchs, D. E. Worrall, V. Fischer, and M. Welling. SE(3)-transformers: 3D roto-translation equivariant attention networks. In *NeurIPS*, 2020.
- [10] M. Geiger and T. Smidt. e3nn: Euclidean neural networks. *arXiv:2207.09453*, 2022.
- [11] N. Gruver, M. Finzi, M. Goldblum, and A. G. Wilson. The Lie derivative for measuring learned equivariance. In *ICLR*, 2023.
- [12] M. Janner, Y. Du, J. B. Tenenbaum, and S. Levine. Planning with diffusion for flexible behavior synthesis. In *ICML*, 2022.
- [13] S.-O. Kaba, A. K. Mondal, Y. Zhang, Y. Bengio, and S. Ravanbakhsh. Equivariance with learned canonicalization functions. In *ICML*, 2023.
- [14] J. J. Kuffner and S. M. LaValle. RRT-connect: An efficient approach to single-query path planning. In *ICRA*, 2000.
- [15] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. In *ICLR*, 2023.
- [16] X. Liu, C. Gong, and Q. Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow. In *ICLR*, 2023.
- [17] J. Milnor. Analytic proofs of the “hairy ball theorem” and the Brouwer fixed point theorem. *American Mathematical Monthly*, 85(7):521–524, 1978.
- [18] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu. WaveNet: A generative model for raw audio. *arXiv:1609.03499*, 2016.
- [19] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI*, 2018.
- [20] O. Puny, M. Atzmon, H. Ben-Hamu, I. Misra, A. Grover, E. J. Smith, and Y. Lipman. Frame averaging for invariant and equivariant network design. In *ICLR*, 2022.
- [21] C. R. Qi, H. Su, K. Mo, and L. J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *CVPR*, 2017.
- [22] A. H. Qureshi, A. Simeonov, M. J. Bency, and M. C. Yip. Motion planning networks. In *ICRA*, 2019.
- [23] H. Ryu, H.-i. Lee, J.-H. Lee, and J. Choi. Equivariant descriptor fields: SE(3)-equivariant energy-based models for end-to-end visual robotic manipulation learning. In *ICLR*, 2023.
- [24] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel. Motion planning with sequential convex optimization and convex collision checking. *IJRR*, 33(9):1251–1270, 2014.
- [25] A. Simeonov, Y. Du, A. Tagliasacchi, J. B. Tenenbaum, A. Rodriguez, P. Agrawal, and V. Sitzmann. Neural descriptor fields: SE(3)-equivariant object representations for manipulation. In *ICRA*, 2022.
- [26] N. Thomas, T. Smidt, S. Kearnes, L. Yang, L. Li, K. Kohlhoff, and P. Riley. Tensor field networks: Rotation- and translation-equivariant neural networks for 3D point clouds. *arXiv:1802.08219*, 2018.
- [27] J. Urain, N. Funk, J. Peters, and G. Chelvatzaki. SE(3)-DiffusionFields: Learning smooth cost functions for joint grasp and motion optimization through diffusion. In *ICRA*, 2023.
- [28] D. Wang, S. Hart, D. Surovik, T. Kelestemur, H. Huang, H. Zhao, M. Yeatman, J. Wang, R. Walters, and R. Platt. Equivariant diffusion policy. In *CoRL*, 2024.
- [29] J. Yang, Z. Cao, C. Deng, R. Antonova, S. Song, and J. Bohg. EquiBot: SIM(3)-equivariant diffusion policy for generalizable and data efficient learning. In *CoRL*, 2024.
- [30] M. Zucker, N. Ratliff, A. D. Dragan, M. Pivtoraiko, M. Klingensmith, C. M. Dellin, J. A. Bagnell, and S. S. Srinivasa. CHOMP: Covariant Hamiltonian optimization for motion planning. *IJRR*, 32(9-10):1164–1193, 2013.